

Design of Bifunctional Fentanyl Analogs: Dual-targeting the Mu-Opioid and Nociceptin Receptors for Safer Analgesia

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Research Problem

PROBLEM

MOR Targeting Compounds carry heavy side-effects

Designing entirely new compounds is slow and high-risk

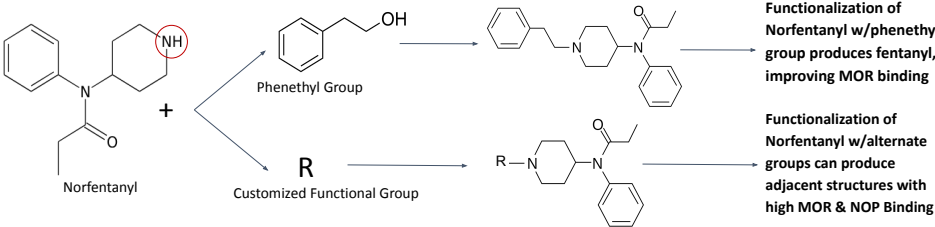
Nociceptin/Orphanin FQ receptor (NOP)

Modification of Fentanyl—Preserves validated, optimal scaffold

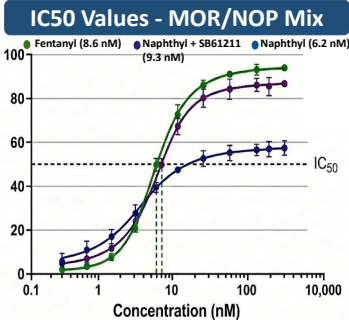
Studies show NOP regulates the same systems that MOR overstimulate. Dual-activation reduces side-effects

Keeping fentanyl structure can preserve pharmacokinetics while selectively altering NOP activation

Proposed Solution

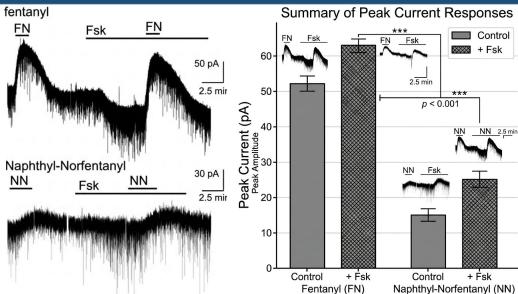


Data Analysis and Results



cAMP assay demonstrated that when Naphthyl-Norfentanyl engages with NOP, MOR activity is significantly reduced

GIRK Electrophysiology



GIRK Shows 2x less harmful conductivity in mouse-derived neurons treated with Naphthyl-Norfentanyl than Fentanyl

Methods

Computational Design of Modified Opioid

Synthesis of Novel Analog Opioid

Biological Validation of Novel Opioid

Use Computational chemistry to discover Fentanyl-Analogs with reduced side-effect potential

Synthesize of optimal Fentanyl Analog Candidate via reductive amination reaction

Test optimal Fentanyl Analog in-vitro to assess binding and signalling activity

Interpretations and Conclusions

Fentanyl

Designed Compound

Potent MOR Activity

High GIRK Signalling

Dangerous and Abusive

55% Emax

Mediated by NOP

Severely Reduced Risks

Developed an improved compound for potent analgesia without side-effects that are overly-severe

Implications

By improving key properties, this compound opens up new pathways for analgesia

(i) Enhances pain treatment as a potent, low-risk alternative to traditional opioids

(ii) Available as a safe prescription without constant medical monitoring.